

Max Elinson

Brooklyn, NY / Toronto, ON || elinsooon.github.io || github.com/elinsooon || max.elinson@gmail.com

EDUCATION

University of Toronto

HBSc in Statistics, Computer Science, and Mathematics

Toronto, ON

June 2026 (Expected)

RELEVANT WORK EXPERIENCE

Department of Fisheries and Oceans

Data Solutions Engineer

Nanaimo, BC & Remote

May 2025 - Present

- Built Python and R ETL pipelines across 7 source datasets / 500,000 records (Oracle, Excel, PostgreSQL), standardizing schema validation and publishing analysis-ready tables for scientists, analysts, and regional managers.
- Produced official maps, figures, and dashboards that helped scientists and managers interpret Pacific salmon stock trends, respond to data requests, and prepare management materials.
- Analyzed 40 years of salmon migration data to identify lower-value sampling periods, projecting a 10% reduction in survey effort without reducing predictive precision.

Ninepoint Partners LP

Software Engineering Intern

Toronto, ON

May 2023 - August 2023

- Built a C#/.NET Core and JavaScript web application with Umbraco CMS integration, enabling non-technical staff to update site content without developer support.
- Developed tax calculators that increased user clicks by 10%, and built an external API data pipeline that reduced manual reporting work for analytics workflows.
- Implemented automated contact routing that reduced response times by 30%, and created reusable HTML/CSS components to standardize new page development.

NYU Langone Health

Student Research Intern

New York, NY

May 2022 - June 2022

- Built R modules to simulate realistic study data with dummy variables, Markov chains, and spline curves, enabling researchers to test analysis workflows before real data collection.
- Refactored simulation utilities to improve runtime efficiency by 20%, allowing larger synthetic datasets and faster parameter testing.

PROJECTS

Toronto Development Timeline Map [\[link\]](#), April 2026

- Built an interactive 3D MapLibre/deck.gl timeline that lets users inspect how Toronto development applications change the city model over time.
- Designed a Python geospatial pipeline to join applications with building footprints, classify status, generate as-built layers, and audit whether completed buildings correctly replace prior massing.

Toronto Traffic Heatmap [\[link\]](#), April 2026

- Built a browser-based traffic visualization that turns tabular Toronto turning movement counts into readable hourly movement patterns across cars, transit, pedestrians, and bikes.
- Processed TMC observations with an OpenStreetMap road graph to generate directed segment flows and hourly route pools, enabling smoother particle animation with Web Worker loading.

Spatial SIR Model with Individual Mobility [\[link\]](#), March 2025

- Built a hub-based stochastic SIR model to study how individual mobility structure changes outbreak-size distributions.
- Used state-space lumping and Monte Carlo simulation in Python to reduce model complexity and identify high-risk hubs.

TECHNICAL SKILLS

- **Languages:** Python, R, SQL, JavaScript, HTML/CSS, C#, Java, C++, C
- **Technologies:** Pandas, GeoPandas, MapLibre GL, PostgreSQL, SQLite, ArcGIS, PowerBI, Django, .NET Core, Git, L^AT_EX